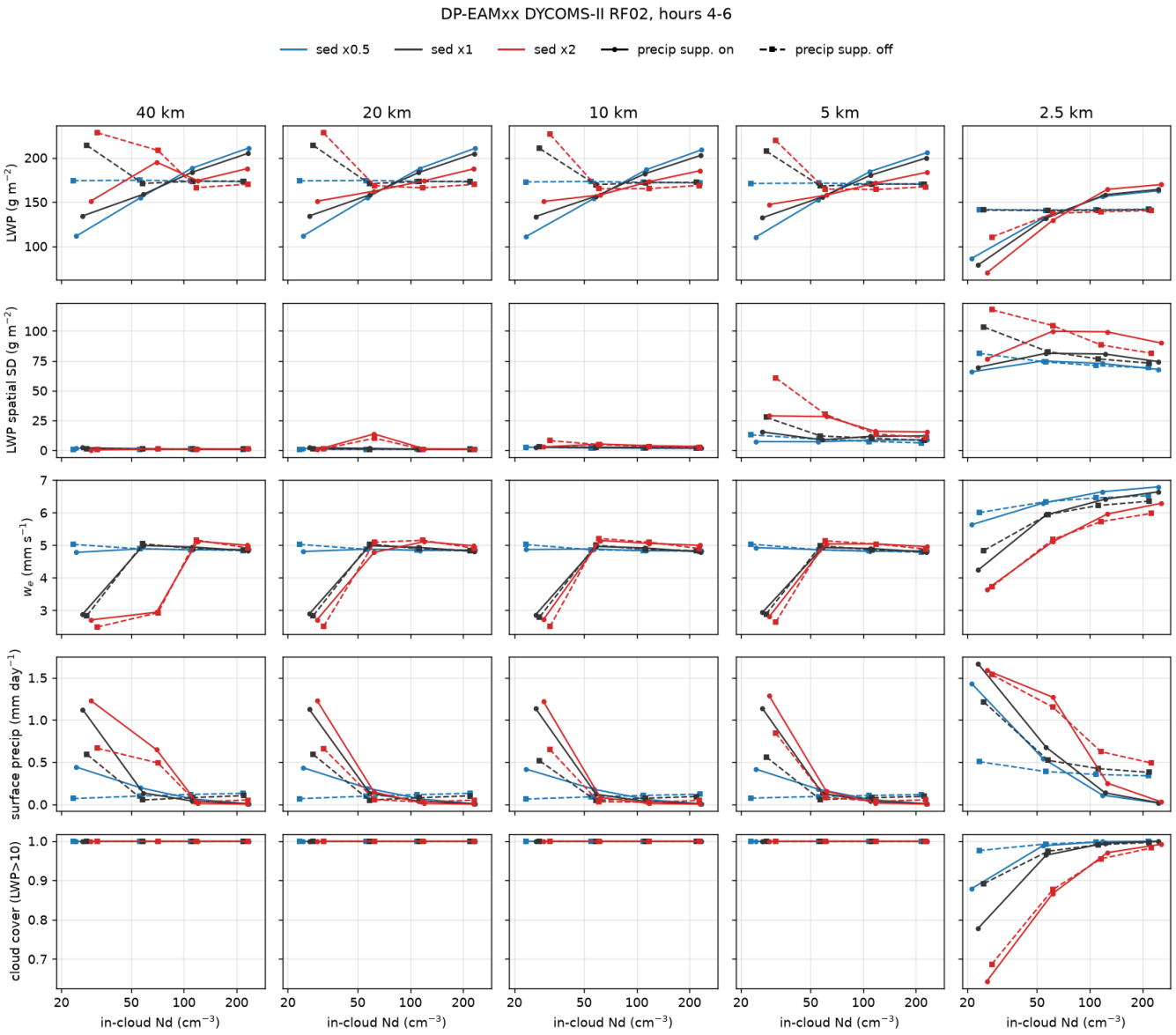
Sed × precip suppression: first results (40–2.5 km)

6 Nov 2026. Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz. DP-EAMxx DYCOMS-II RF02, 6 h, hour 4–6 means. These are 120 of the 144 runs; the 1.25 km runs are queued and sub-km is on hold. These are single realizations.



What the runs show

1. **Consistency check.** The sed×1, suppression-on cells reproduce the earlier ACI-study numbers to within ∼1–2 %.
2. **At ≥5 km (single-column-like), the positive LWP–Nd response is all precip suppression.** With suppression off, LWP is flat at 170–175 g m^{-2} ($S \approx 0$), wherever w_e stays near 4.9 mm s^{-1}. With suppression on, $S_{\text{high}} \approx 0.15$.
3. **Sedimentation sets an entrainment regime at coarse resolution.** At Nd 20 (sed×1, ×2), and at Nd 50 with sed×2 at 40 km, w_e drops from ∼4.9 to ∼2.5–2.9 mm s^{-1}. This happens with suppression on or off. With suppression off, LWP then rises to 210–230 g m^{-2}, which gives a *negative* low-Nd branch (S_{low} from −0.3 to −0.5). Sed×0.5 removes this regime entirely. This looks like the sedimentation–entrainment feedback, acting as a threshold in the single-column limit.

4. **At 2.5 km**, with suppression off and sed×0.5 or ×1, LWP is again flat ($\approx$141 g m^{-2}), even though w_e still rises with Nd. With suppression on, S_{low} grows with sed (0.45, 0.57, 0.71 for ×0.5, ×1, ×2), and $S_{\mathrm{high}} \approx 0.05$ for every sed value. Sed×2 lowers w_e by 0.3–1.1 mm s^{-1} at every Nd. It also breaks up the low-Nd deck even with suppression off (cloud cover 0.64–0.69 at Nd 20).
5. **LWP spatial SD** is 1–3 g m^{-2} at $\geq$10 km, 6–60 at 5 km and 65–120 at 2.5 km. It grows with the sed factor and is largest at low Nd. The online value (expression `mean('col')` of the squared anomaly) matches the SD computed afterwards from the 2D LWP output.
6. **AeroCom cloud top (PR #9017):** liquid cloud-top CDNC is $\approx$0.7× the in-cloud median Nd, consistent with SPA CCN decreasing upward. r_{eff} at cloud top ranges from 8 to 21 μm and decreases with Nd and with the sed factor.

Notes. Surface precipitation includes the cloud-droplet sedimentation flux. With suppression off, the autoconversion rate matches the default at Nd 63, so rain is higher than default above that Nd and lower below it. Susceptibility $S = d\ln\mathrm{LWP}/d\ln N_d$ uses the diagnosed Nd; the full table is in `susceptibility.csv`. **Outputs:** `$PSCRATCH/dp_screamxx_sedps/analysis/` (`summary_h4-6.csv`, `timeseries.nc`).
Code: E3SM `mahf708/eamxx/p3-cloud-sed-factor` and scmlib `sed-precsupp-study`, both on mahf708's forks.